# Olivier Côté ACAS

PhD Candidate in Actuarial Science
Université Laval

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LANGUAGES Français (native), English (C1), Svenska (A1)

## Education

**PhD in Actuarial Science (GPA 4.17/4.33)** · 2023–Expected 2027
**Université Laval**
Fairness in insurance pricing, supervised by Prof. Marie-Pier Côté (Université Laval) and Prof. Arthur Charpentier (UQÀM).

**Master's degree in Actuarial Science, research-based (GPA 4.24/4.33)**, Université Laval · 2021–2023
**Bachelor's degree in Actuarial Science (GPA 4.17/4.33)**, Université Laval · 2018–2021

**Research visits** · 2023–2026
- Department of Mathematics, Stockholm University (2026): seminar and discussions with Prof. Mathias Millberg Lindholm.
- Department of Statistics, University of Toronto (2025): seminar and discussions with Prof. Christopher Blier-Wong.
- Bayes Business School, London (2023): discussions with Prof. Andreas Tsanakas.

## Research papers

**Côté, O.**, Côté, M.-P., and Charpentier, A. (2027). A statistical perspective on fairness in insurance pricing. *Annual Review of Statistics and Its Application*. In press

Charpentier, A., **Côté, O.**, and Côté, M.-P. (2026b). Governing fairness in insurance pricing: The trilemma of actuarial adequacy, solidarity, and causal legitimacy. *Journal of Business Ethics*. doi:10.1007/s10551-026-06479-2

Côté, M.-P., **Côté, O.**, and Charpentier, A. (2026). Recoverability of market-wide fair insurance premiums under selection bias. *Insurance: Mathematics and Economics*, 129:103268. doi:10.1016/j.insmatheco.2026.103268

**Côté, O.**, Côté, M.-P., and Charpentier, A. (2026). A scalable toolbox for exposing indirect discrimination in insurance rates. *CAS Research Paper Series on Bias and Insurance*. Research paper, casact.org
- Winner of the American Academy of Actuaries Award for Research 2026
- Selected by the National Association of Insurance Commissioners (NAIC) for its webinar series for U.S. regulators

Charpentier, A., Fernandes Machado, A., **Côté, O.**, and Hu, F. (2026a). Federated measurement of demographic disparities from quantile sketches. *arXiv:2602.18870*. Preprint

**Côté, O.**, Côté, M.-P., and Charpentier, A. (2025). A fair price to pay: Exploiting causal graphs for fairness in insurance. *Journal of Risk and Insurance*, 92(1):33–75. doi:10.1111/jori.12503
- Finalist for the American Academy of Actuaries Award for Research 2025

## Diversity, equity and inclusion education

**Get Comfortable Being Uncomfortable: Dialogue About Favoritism and Fairness**, CANSSI · 2025
**Let's talk about race and collective responsibility**, CANSSI · 2024
**Acting as a sentinel for the prevention of suicide**, Centre de prévention du suicide du Québec · 2023
**Addressing Conflicts Related to Bias, Privilege, and Identity in the STEM Fields**, CANSSI · 2022

## Professional designations

**Associate of the Casualty Actuarial Society (ACAS)**, Casualty Actuarial Society · 2025
**Professionalism Workshop**, Canadian Institute of Actuaries · 2025
**MAS-I, MAS-II, Exams 5 and 6C, Online Courses 1 and 2**, CAS · 2020–2025
**Exams P, FM and IFM**, Society of Actuaries · 2019–2020

## Research affiliations

**Institut Intelligence et Données, Big Data Research Center**, Université Laval · 2021–present
**Quantact research laboratory, Centre de recherches mathématiques**, Montréal · 2021–present

## Journal refereeing

***European Actuarial Journal*** · 2026–present
***Annals of Actuarial Science*** · 2026–present
***Journal of Risk and Insurance*** · 2024–present

## Scholarships and awards

**Award for Research** 2026
**American Academy of Actuaries**
Winner for *A scalable toolbox for exposing indirect discrimination in insurance rates*. Annual international competition for papers by early-career scholars. Honorarium US$7 500 (≈ **75 000 SEK**).

**Travel grants** 2026
**NSERC, American Academy of Actuaries, Big Data Research Center**
Three competitive grants for conference travel: NSERC General Graduate Studies Fund ($1 300 CAD, ≈ **9 100 SEK**), American Academy of Actuaries (US$1 500, ≈ **15 000 SEK**) and Big Data Research Center ($2 000 CAD, ≈ **14 000 SEK**).

**Canada Graduate Research Scholarship – Doctoral** 2025–2028
**Natural Sciences and Engineering Research Council of Canada**
National competition, held from May 2025. $120 000 CAD over three years (≈ **841 000 SEK**).

**Hickman Scholars Program** 2024–2027
**Society of Actuaries**
International competition. The three renewals were competitive. US$20 000 per year (≈ **199 000 SEK**).

**Doctoral Training Scholarships** 2024–2025
**Fonds de recherche du Québec – Nature et technologies**
Provincial competition, held from May 2024 to May 2025, when the NSERC doctoral scholarship began. $100 000 CAD offered over four years (≈ **701 000 SEK**).

**Mitacs Accelerate** 2024–2027
Research partnership with a large Canadian P&C insurer, in which I am the lead student researcher. Selected after a scientific peer review and an internal Mitacs review. $84 000 CAD over three years (≈ **589 000 SEK**).

**Doctorate attraction grant Florent Toureille** 2023–2026
**Chaire d'actuariat de l'Université Laval**
Institutional competition based on leadership and excellence. $30 000 CAD over three years (≈ **210 000 SEK**).

**Leadership and Engagement Scholarship** 2023–2026
**Université Laval**
Institutional competition in the sports sector. $30 000 CAD over three years (≈ **210 000 SEK**).

**Charles Brindamour Graduate Scholarship for Excellence** 2023
**Université Laval, funded by Charles Brindamour (CEO of Intact Financial Corporation)**
Institutional competition based on excellence, one award a year. $10 000 CAD (≈ **70 000 SEK**).

**François-Laviolette Scholarship** 2022
**Université Laval**
Institutional competition for responsible artificial intelligence. $7 500 CAD (≈ **53 000 SEK**).

**Grant of attraction to the Master's** 2021
**Chaire d'actuariat de l'Université Laval**
Institutional competition recognizing research potential. $15 000 CAD (≈ **105 000 SEK**).

**CAS Trust Scholarship** 2021
**Casualty Actuarial Society**
International competition. US$2 500 (≈ **25 000 SEK**).

**Alexander Graham Bell Canada Graduate Scholarship – Master's** 2021–2022
**Natural Sciences and Engineering Research Council of Canada**
National competition. $17 500 CAD (≈ **123 000 SEK**).

**Lieutenant Governor's Youth Medal** 2018
**Cégep de Lévis**
Awarded for outstanding involvement and academic results, on nomination by the teaching staff.

## Supervision

**Co-supervisor of a master's student, Université Laval** 2025–2026
**With Prof. Marie-Pier Côté, Mitacs Accelerate partnership**
Participated in negotiating the student's Mitacs funding with the industrial partner and in designing the project. The student continued to the PhD through the fast track.

**Co-supervisor of a master's student, Université Laval** 2023
**With Prof. Marie-Pier Côté**
One-year project on fairness in insurance. The student now works as an actuary.

## Committees

**Institut intelligence et données, Université Laval**, board member — 2025–present
**Graduate Program Committee in Actuarial Science, Université Laval**, voting member — 2022–present
**Big Data Research Center, Université Laval**, student voting member — 2022–2025
**Actuarial Science Student Investment Fund, Université Laval**, voting member — 2021–2024

## Event organization

**Workshop on fairness and discrimination in insurance** — 2026
**Université Laval, 6–7 October 2026, 90 participants**
Member of the organizing committee, host of the second day, panel moderator and presenter.

**Quantact Student Day** — 2025
**Province-wide event, 3 October 2025, 40 students**
Organizer: recruited a five-member committee, and coordinated seven presenters.

## Work experience

**Online course on fair insurance pricing, project lead and script writer** — 2026
**Casualty Actuarial Society (English and French versions, in post-production)**
Commissioned by the CAS. Coordinated four parties: the university team (authors and research assistants), the CAS task force on biases in insurance pricing, the CAS educational development team and a production company. Two hours of two-minute videos, activities and quizzes, co-presented with M.-P. Côté and A. Charpentier.

**Course lecturer, two mandatory undergraduate courses (3 credits each)** — 2022–2023
**Université Laval**

- Linear Models in Actuarial Science (fall 2022, 88 students)
- Statistical Learning in Actuarial Science (winter 2023, 54 students)

*Awards:* Faculty of Science and Engineering Teaching Star, one per course (fall 2022 and winter 2023).

**American football coach** — 2019–2025
**Falcons of Cégep de Lévis**
American football coach for different teams for more than ten years (2012–2025), including seven years in collegiate football at the Cégep de Lévis (2019–2025):

- Tight ends and Fullbacks coach (2019–2025), responsible for the development of 10 athletes
- Supervisor for a bachelor's degree intern in sports intervention (2023)
- Volunteer mathematics tutor for student-athletes

*Awards:* RSEQ Coaching Staff of the Year 2023 and 2024, Sportsmanship Award 2022 (D1) and 2023 (D2).

**Teaching assistant** — 2019–2024
**Université Laval**
Ten times: probability $\times 3$, statistical learning $\times 5$, linear models and internship evaluation.

**Actuarial intern** — 2019–2021
**Desjardins General Insurance Group**
Four internships: property ratemaking (2019), business analysis (2020 and 2021) and research (2021).

## Academic communications

**Operationalizing Fairness in Actuarial Pricing: From Principle to Practice** — 2025–2026
**Casualty Actuarial Society and University of Toronto**
Sole presenter at a 90-minute CAS webinar (2026, 950 registrants). Presented at the CAS Annual Meeting in Austin (2025, over 150 participants) and at the CAS Canada Connect Day in Toronto (2025, over 100). Seminar at the Department of Statistics, University of Toronto (2025). *Audience: practitioners and academics.*

**A Scalable Toolbox for Exposing Indirect Discrimination in Insurance Rates** — 2025–2026
**International conferences and invited talks**
Presented at the first ASTIN Bulletin Conference in Zurich (2026) and at PARTY in Liverpool (2025, 60 doctoral students from 29 countries). Presented in French with Arthur Charpentier at the Chaire ACTIONS and Milliman R&D in Paris (2025, 80 participants). *Audience: academics and practitioners.*

**Profiling Actuarial Discrimination via Causal Decomposition** — 2026
**Conferences and seminars**
Talk at the 29th International Congress on Insurance: Mathematics and Economics in Seoul and poster at the Actuarial and Financial Mathematics Conference in Brussels. Seminar at STATQAM, UQÀM, also covering confounding bias, and research talk at the Big Data Research Center, Université Laval. *Audience: academics.*

**The Trilemma of Fairness in P&C Pricing** 2026
**American Academy of Actuaries and AAIARD**
Invited session of the American Academy of Actuaries at the 61st Actuarial Research Conference. Presented in French to the Association des actuaires en IARD du Québec as *Le trilemme de l'équité en tarification IARD*. *Audience: decision-makers and practitioners.*

**Bias in Assessing Financial Risk: Insights from the 2025 Research Award** 2026
**American Academy of Actuaries**
Invited panelist as a 2025 finalist. Webinar to 445 actuaries and practitioners. *Audience: practitioners.*

**A Causal Perspective on Fairness in Insurance Pricing** 2026
**Department of Mathematics, Stockholm University**
Seminar during research stay. *Audience: academics.*

**Two invited presentations at the 60th Actuarial Research Conference** 2025
**American Academy of Actuaries panel & Casualty Actuarial Society session, Toronto**
Guest panelist and presenter with Marie-Pier Côté and Arthur Charpentier. *Audience: decision-makers.*

**A Fair Price to Pay: Exploiting Causal Graphs for Fairness in Insurance** 2024
**Conference and workshop**
Talk at the 27th International Congress on Insurance: Mathematics and Economics in Chicago. Presented to over 60 academics, practitioners and regulators at the Workshop on fairness and discrimination in insurance. *Audience: academics.*

**Perspective causale sur la discrimination indirecte en assurance générale** 2024
**Autorité des marchés financiers**
Presentation to decision-makers at Québec's financial authority, followed by interviews with ten regulators on their needs. *Audience: decision-makers.*

**Perspective causale sur les discriminations dans les modèles prédictifs en assurance** 2024
**Institut intelligence et données**
Guest presenter to over 100 people from academia. *Audience: academics.*

**Équité et discrimination en assurance générale** 2023
**Intact Financial Corporation**
Presentation to the CEO, CLO and CDO. *Audience: decision-makers.*

**Causal Inference and Fairness in Insurance Pricing** 2023
**Insurance Data Science Conference, London**
Conference presentation. *Audience: academics.*

## Public outreach

**L'avenir à travers une boule de cristal mathématique : Le travail d'un actuaire** 2023–2026
**Cégep de Lévis**
Guest presenter for CEGEP students. *Audience: students.*

**Roundtable on raising awareness about youth bullying** 2025
**Mammouth hebdo - Télé-Québec**
Québec television roundtable. *Audience: general public.*

**Pour une intelligence artificielle équitable en assurance** 2024
**Université Laval News**
I collaborated for a scientific communication article. *Audience: general public.*

**Perspective causale sur la discrimination indirecte en assurance générale** 2024
**Institut Intelligence et données**
4th research competition on data intelligence. ≈ 50 graduate students. *Audience: general public.*

**Perspective causale sur la discrimination indirecte en assurance générale** 2024
**Institut de valorisation des données**
I presented my research during IVADO Digital Future Day 2024. Over 250 people. *Audience: general public.*

**L'IA pour calculer le prix de votre assurance auto: poétique ou pas éthique?** 2023
**Big Data Research Center**
Presentation during the event *lightning talks 180*. *Audience: general public.*
*Awards:* Public's choice award

**Artificial intelligence to prevent unfair discrimination in auto insurance** 2023
**IVADO**
I authored an 800-word scientific communication article. *Audience: general public.*